

23	A heater of resistance $5.0\ \Omega$ dissipates $125\ \text{W}$ of power when current flows through it. The amount of energy converted to heat when $0.4\ \text{C}$ of charge passes through it is							
	A	2 J	B	10 J	C	50 J	D	625 J
Answer: B $P = \frac{V^2}{R} \Rightarrow V^2 = 125 \times 5 \Rightarrow p.d\ across\ heater, V = \sqrt{625} = 25\ \text{V}$ Since $V = \frac{W}{Q} \Rightarrow \text{Amount of energy converted to heat, } W = VQ = 25 \times 0.4 = 10\ \text{J}$								

